

CA1: Individual Requirements and Domain Analysis

Weighting and schedule 15% of the module. Individual work using a unique student scenario.

1 Purpose

CA1 assesses the student's ability to transform a short, incomplete scenario into a structured and defensible requirements and domain analysis. The scenario deliberately contains ambiguity, implied needs and missing information. Students must distinguish evidence, assumptions and questions rather than inventing certainty.

2 Relationship to teaching

Teaching covered before completion	How it appears in CA1
Business problems, stakeholders, purpose and scope	Problem statement, stakeholder catalogue, boundary, assumptions, constraints and risks.
Functional and non-functional requirements, business rules and MoSCoW	Requirements catalogue, rule register, measurable qualities and prioritisation rationale.
User stories, acceptance criteria, Gherkin, actors and use cases	Actor-goal catalogue, use-case model, detailed use case and behavioural scenarios.
Candidate concepts and domain vocabulary	Candidate extraction, classification, glossary and justified refinement.
Domain modelling, associations, attributes and multiplicity	Conceptual UML domain model and modelling rationale.

3 Required submission

Submit one integrated analysis dossier of approximately 2,500–3,500 words, excluding diagrams, tables, references and appendices, together with the individual repository containing editable artefact sources and development history. Submit the final format and repository link through Canvas as directed.

4 Tools and how to use them

Tool	Expected use	Evidence to submit
Git and GitHub	Create an individual private repository at the start. Commit the developing analysis at meaningful milestones rather than uploading everything at the end.	Repository link, README, readable history and tagged CA1 release.
Word or Markdown	Develop the integrated dossier with stable identifiers and consistent terminology.	Editable report and PDF where required.
PlantUML	Create the use-case diagram and conceptual domain model. Do not include methods, software services, database design or navigability in the domain model.	The .puml source and readable exported image for each diagram.
Gherkin	Write three to five concrete acceptance examples using domain language. Include success, alternative/failure and boundary/rule behaviour.	One or more .feature files linked to requirements and use cases.
Traceability table	Maintain clear links between problem evidence, stakeholder goals, requirements, scenarios, use cases and concepts.	Embedded table in the report and editable source where separate.

5 Required CA1 artefacts

Artefact	What it must include	Quality checks
Scenario interpretation	Organisation, present process, problems, consequences, desired outcomes, known evidence and uncertainty.	Technology is not presented as the problem; unsupported claims are marked as assumptions or questions.
Purpose, scope and boundary	Concise purpose; in-scope and out-of-scope functions; dependencies, assumptions and constraints.	Scope is realistic and consistent with the problem.
Stakeholder and actor catalogue	Direct, indirect and external stakeholders; goals, concerns and influence; primary/supporting actors and offstage stakeholders; actor-goal table.	Actors cross the system boundary; offstage stakeholders still influence requirements.
Requirements catalogue	Approximately 8–12 functional requirements, at least 4 measurable non-functional requirements, source/rationale, MoSCoW priority and verification evidence.	Requirements are atomic, clear, feasible, testable, consistent and traceable.
Rules and uncertainty register	At least 4 business rules where supported; assumptions, constraints, risks and at least 5 clarification questions.	Assumptions are not disguised as confirmed rules.
Use-case model	Named system boundary, actors, coherent use-case catalogue and one detailed use case with preconditions, trigger, success flow, alternatives, exceptions and postconditions.	Use cases represent actor goals and agree with the requirements.
Gherkin examples	Three to five scenarios using Feature/Rule where useful and Given/When/Then steps.	Observable outcomes; no screen, database or programming detail.
Candidate-concept analysis	Candidate terms, classification, keep/merge/convert/remove decisions, evidence and glossary.	Reasoning is visible rather than only a final class list.
Domain model	Approximately 8–15 conceptual classes, selected attributes, named associations and multiplicity.	No methods, controllers, repositories, UI controls, foreign keys or design navigation.
Traceability matrix	Principal chains from problem/stakeholder goal to requirement, Gherkin/use case and domain concept.	No major Must Have requirement is isolated.
Reflection and AI declaration	Two or three important decisions, revisions, limitations, unresolved questions and transparent AI disclosure.	Student can explain every submitted artefact.

6 User stories, acceptance criteria and storage in CA1

CA1 produces candidate product information rather than a team Scrum backlog. User stories are used as a stakeholder-value view of selected requirements. They do not replace the requirements catalogue or detailed use cases. The student should store stories and acceptance criteria in a version-controlled backlog file or structured appendix, and link concrete examples to Gherkin feature files.

Record	Required content	Authoritative location
User story	Stable ID; short title; As a / I want / so that statement; business value; priority; related stakeholder and requirement IDs.	Repository backlog file or structured report appendix.
Acceptance criteria	Observable conditions, important rules, constraints and outcomes that indicate acceptable completion.	Stored with the story, with stable identifiers where useful.
Gherkin examples	Concrete data and context illustrating success, rejection, exception and boundary behaviour.	Version-controlled .feature files.
Formal requirement	Atomic requirement, source/rationale, type, priority, verification method and traceability.	Requirements catalogue in the dossier/repository.

Record	Required content	Authoritative location
Use case	Actor goal, trigger, preconditions, flows and postconditions.	Use-case section and PlantUML diagram source.

Avoid duplication The same information should not be copied independently into Word, Trello and GitHub. Store the authoritative item once, then link to it from other tools. Trello or Planner becomes the workflow view in CA2.

6.1 Recommended user-story template

- Story ID and short title.
- User story statement: As a [role], I want [capability], so that [value].
- Business value and related stakeholder goal.
- Related requirement, business-rule and use-case identifiers.
- MoSCoW priority.
- Concise acceptance criteria.
- Related Gherkin scenario identifiers.
- Questions, assumptions or dependencies.

7 Recommended repository structure

```
ca1-analysis/
  README.md
  report/
  requirements/
  features/
  diagrams/
    use-case-model.puml
    domain-model.puml
  decisions/
  references/
```

8 CA1 checkpoints

Week	Checkpoint	Minimum evidence
2	Scenario interpretation	Problem statement, initial stakeholders, assumptions, questions and first repository commits.
3	Requirements and behaviour	Draft requirements, actor-goal catalogue, use-case model and initial Gherkin examples.
4	Domain analysis and walkthrough	Candidate analysis, glossary, draft domain model, traceability and brief individual explanation.

9 CA1 marking rubric

Criterion	Weight	Evidence expected
Problem, purpose, scope and uncertainty	15%	Clear problem evidence; defensible boundary; relevant assumptions, constraints, risks and questions.
Stakeholders, actors and goals	15%	Complete classification and clear connection between interests, goals and system needs.

Criterion	Weight	Evidence expected
Requirements and business rules	25%	Clear, measurable, prioritised and traceable requirements and rules.
Use cases and behavioural examples	20%	Correct use-case model, detailed flow and meaningful Gherkin success, alternative and boundary examples.
Candidate concepts and domain model	20%	Justified concepts, glossary, associations, multiplicity and attributes; model remains conceptual.
Rationale, traceability, professional evidence and AI disclosure	5%	Decisions are explained; repository shows progress; sources and AI use are transparent.

10 AI Level 3 guidance for CA1

CA1 assesses individual interpretation and analysis. Students should create an initial evidence-based analysis before using AI for critique or refinement. AI suggestions must be distinguishable from scenario evidence and must not be used to manufacture certainty where the scenario is ambiguous.

Appropriate CA1 use	Required student action
Generate possible stakeholder questions or identify overlooked ambiguities.	Compare with the scenario, retain only relevant questions and explain why each matters.
Review requirement clarity, atomicity, measurability or testability.	Apply taught quality criteria and make the final wording decision.
Suggest alternative acceptance criteria or Gherkin examples.	Verify against requirements and business rules; adapt examples to the domain.
Critique a student-created use-case or domain model.	Check all suggestions against evidence and UML analysis rules; reject unjustified software/design concepts.
Review traceability for gaps.	Confirm links manually and explain unresolved gaps.
Improve structure or language.	Preserve the student's technical meaning and own voice.

CA1 boundary AI must not generate the complete scenario interpretation, requirements catalogue, stakeholder analysis, candidate-concept decisions, domain model or personal reflection. The student must be able to defend the origin and rationale of every item.

10.1 CA1 AI artefacts

- AI declaration in the analysis dossier.
- Representative interaction log in the repository or appendix.
- At least one clear example of an AI suggestion that was checked, modified or rejected, where AI was used.
- Git history showing the student's own progressive analysis.

11 What happens after CA1

CA1 is graded individually. In CA2, each group compares the members' CA1 scenarios using a suitability matrix. The selected scenario is approved by the lecturer and normally receives an expansion brief. The group must reconcile and extend the analysis rather than simply resubmit one student's CA1 work.